

Auteur: Abdellah El Moussaoui

Studierichting: Informatica

Oefeningenreeks 2D nr. 6.2

$$f(x) = \frac{x^2 - 5x + 6}{x^3 - 2x^2 - x + 2} = \frac{(x-2)(x-3)}{x(x^2-1) - 2(x^2-1)} = \frac{(x-2)(x-3)}{(x^2-1)(x-2)} = \frac{x-3}{x^2-1} \text{ voor } x \neq 2$$

- $\lim_{x \rightarrow 2} f(x)$

$$\lim_{x \rightarrow 2} \frac{x-3}{x^2-1} = \frac{-1}{3}$$

- $\lim_{x \rightarrow 3} f(x)$

$$\lim_{x \rightarrow 3} \frac{x-3}{x^2-1} = 0$$

- $\lim_{x \rightarrow -1} f(x)$

$$\lim_{x \rightarrow -1+} \frac{x-3}{x^2-1} = \frac{x-3}{(x-1)(x+1)} = +\infty$$

$$\lim_{x \rightarrow -1-} \frac{x-3}{x^2-1} = \frac{x-3}{(x-1)(x+1)} = -\infty$$

- $\lim_{x \rightarrow 1} f(x)$

$$\lim_{x \rightarrow 1+} \frac{x-3}{x^2-1} = \frac{x-3}{(x-1)(x+1)} = -\infty$$

$$\lim_{x \rightarrow 1-} \frac{x-3}{x^2-1} = \frac{x-3}{(x-1)(x+1)} = +\infty$$

- $\lim_{x \rightarrow \infty} f(x)$

$$\lim_{x \rightarrow \infty} \frac{x-3}{x^2-1} = 0$$

References